From papers to a TACO research agent

Paper2Agent reading + CAUSEWAY implementation proposal

- Meeting discussion • approximately 10 minutes
- Goal: make research methods easier to inspect and use
- Separate published findings, working software and future experiments

Paper: https://arxiv.org/abs/2509.06917v2

What does Paper2Agent contribute?

Paper summary • preprint v2, October 2025

- Connect a paper, its code and data to a conversational assistant.
- Expose reusable functions through Model Context Protocol (MCP).
- Test functions against reference examples before exposing them.

Paper2Agent v2, overview and extended methods: https://arxiv.org/html/2509.06917v2

How the paper builds an agent

Paper summary • the reusable workflow

- Locate code → configure environment → inspect tutorials.
- Execute reference examples → extract tools → validate results.
- Tools perform work; resources provide context; prompts guide workflows.
- Connect the resulting server to an LLM-based assistant.

Paper2Agent v2, overview and extended methods: https://arxiv.org/html/2509.06917v2

Evidence and limits

Paper summary • what the demonstrations establish

- Case studies: AlphaGenome, TISSUE and Scanpy.
- AlphaGenome: 15/15 tutorial queries and 15/15 novel queries correct in the reported benchmark.
- These are selected computational benchmarks, not clinical validation.
- Incomplete source code can prevent reliable tool creation.

Paper2Agent v2, results and discussion: https://arxiv.org/html/2509.06917v2

Apply the idea to CAUSEWAY

Our proposal • not an experiment reported in Paper2Agent

- Researcher asks a question about TACO.
- Assistant selects a documented tool and checks its inputs.
- Tool returns structured evidence with source links and status.
- Assistant explains the result and what remains unknown.

Local design: ARCHITECTURE.md and taco_agent/

First TACO agent: a small useful start

Our implementation scope

- Retrieve TACO purpose, input requirements and limitations.
- Check whether a proposed data manifest is complete.
- Expose tools, a reference resource and a workflow prompt via MCP.
- Prediction execution remains unavailable until the upstream pipeline is reproduced.

Local implementation proposal; upstream: https://github.com/SindyPin/TACO

Patient Portal: add useful context

Requested feature • fictional demo data only

- Comorbidities: other conditions a person has.
- Current medicines: name, dose, frequency and reason for use.
- Distinguish no conditions/medicines from information not supplied.
- Show the same session context to the doctor.

Researcher Portal: compare the methods

Requested feature • keep each task distinct

- Causal discovery: MR, Control Theory and DCE.
- Subtyping: expression-based clustering with symptom interpretation.
- Early detection: TACO, combining selected genes and TabPFN.
- Drug research: SPLIT, using cohort-specific predictions.

How can methods be combined?

Our experiment planning tool

- Select methods and inspect the proposed combination.
- Check shared gene identifiers, cohort definitions and data requirements.
- Label published pipelines separately from proposed combinations.
- Compare methods with task-specific metrics and independent validation.

Doctor Portal: explain what is visible

Requested feature • interpretability

- Show which selected symptoms match each displayed research theme.
- Show the rule, its source and its limitations.
- Keep comorbidities and medicines visible as context.
- Show model explanations only after a model has actually run.

Local design: Doctor Portal explanation panel

Keep the Shiny application in the workflow

Key resource supplied by the project team

- Open the existing Shiny explorer from the Researcher Portal.
- Use the associated paper to understand its MR/CT results.
- Keep external exploration separate from locally executed analyses.
- Live Shiny contents still need checking; the research browser could not open the site.

https://sindypin.shinyapps.io/github/ • associated paper: https://doi.org/10.1371/journal.pcbi.1013725

What the team should decide next

Meeting discussion

- Which TACO commit, preprocessing and model artifact are authoritative?
- Which reference dataset and expected outputs will verify reproduction?
- Which method combinations should be tested first?
- Who will review the scientific interpretation and future patient-data workflow?

Reading and reference links

Versioned paper + project resources

- Paper2Agent preprint v2: arxiv.org/abs/2509.06917v2
- Paper2Agent journal article (2026): doi.org/10.1038/s41586-026-11044-y
- TACO: github.com/SindyPin/TACO
- MR/CT paper: doi.org/10.1371/journal.pcbi.1013725
- Shiny explorer: sindypin.shinyapps.io/github/
- SPLIT: github.com/SindyPin/SPLIT